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Lema integral de Cauchy

Lema 1. *Sea γ un camino y $g\colon \operatorname{Im}(\gamma) \to \mathbb{C}$ continua tal que $|g| \leqslant M$ para algún $M > 0$.*

$$\implies \forall z \in \mathbb{C} \setminus \operatorname{Im}(\gamma) : z \longmapsto G(z) = \int_\gamma \frac{g(w)}{w-z}\,\mathrm{d}w \text{ es holomorfa y } G'(z) = \int_\gamma \frac{g(w)}{(w-z)^2}\,\mathrm{d}w.$$

***Demostración*:** Sea $z_0 \in \mathbb{C} \setminus \operatorname{Im}(\gamma)$, tomamos $D = \operatorname{dist}(z_0, \operatorname{Im}(\gamma)) > 0$ y consideramos $h \in \mathbb{C}$ tal que $|h| < {}^{D}\!/_{2}$. Entonces,

$$\begin{aligned}
\frac{G(z_0+h) - G(z_0)}{h} &= \int_\gamma \left(\frac{1}{w-(z_0+h)} - \frac{1}{w-z_0} \right) \frac{1}{h}\, g(w)\,\mathrm{d}w \\
&= \int_\gamma \frac{1}{(w-z_0-h)(w-z_0)} g(w)\,\mathrm{d}w.
\end{aligned}$$

Por tanto, se satisface que

$$\begin{aligned}
\left| \frac{G(z_0+h) - G(z_0)}{h} - \int_\gamma \frac{g(w)}{(w-z_0)^2}\,\mathrm{d}w \right| &= \left| \int_\gamma g(w) \left(\frac{1}{(w-z_0-h)(w-z_0)} - \frac{1}{(w-z_0)^2} \right) \mathrm{d}w \right| \\
&\leqslant \int_\gamma |g(w)| \left| \frac{(w-z_0)-(w-z_0-h)}{(w-z_0-h)(w-z_0)^2} \right| |\mathrm{d}w| \\
&\leqslant M|h| \int_\gamma \frac{|\mathrm{d}w|}{|(w-z_0-h)(w-z_0)^2|} \\
&\leqslant |h|\, \frac{M}{D^2 \, {}^{D}\!/_{2}} \operatorname{long}(\gamma).
\end{aligned}$$

Por último, al tomar el límite cuando $h \to 0$,

$$\lim_{h\to 0} \left| \frac{G(z_0+h) - G(z_0)}{h} - \int_\gamma \frac{g(w)}{(w-z_0)^2}\,\mathrm{d}w \right| = 0.$$

Es decir, G es derivable en z_0 y su derivada es $G'(z) = \int_\gamma \frac{g(w)}{(w-z_0)^2}\,\mathrm{d}w$. Como z_0 era arbitrario, G es holomorfa en $\mathbb{C} \setminus \operatorname{Im}(\gamma)$. ■

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